

Silver Image Data Collection and Detection

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I. Investigation and Datasets

Silver mining is a vital industrial sector in Northern Idaho; however, accurately identifying native silver deposits remains both challenging and time-consuming. Traditional identification methods rely on manual visual inspection by experienced geologists and laboratory-based analytical techniques. This research compares Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) as solutions for real-time, scalable mineral identification methods using a monolithic image dataset of silver and non-silver images collected from the Mindat website (mindat.org). Mindat aggregates user-contributed photographs from field collectors, private collections, museums, and mining sites, resulting in a diverse dataset that reflects real-world variability in mineral appearance. A total of 20,564 original JPEG images (3,869 silver and 16,695 non-silver), with varying resolutions and pixel configurations, were collected, corresponding to a class distribution of 18.8% silver and 81.2% non-silver. The images were organized into two separate directories and combined to form a binary classification dataset. After filtering and data cleansing, the final dataset was balanced to 3,655 silver and 3,655 non-silver images for training and evaluation, where-for the non-silver class -- 2,559 are “hard negatives” and the remaining 1,096 are “easy negatives”. For this project, “hard negatives” refers to minerals that appear similar to silver and thus are hard to distinguish as non-silver. The term “easy negatives” refers to all the remaining non-silver minerals that are not considered “hard negative”; they are considered easy to distinguish as non-silver. Table I shows all collected metadata fields for the dataset and their descriptions.

Field Name	Description
image_index	Unique integer index assigned to each image
label	Class label (0 = non-silver, 1 = silver)
mineral	Mineral name associated with the image
format	Image file format (ex: JPEG, PNG)
file_size	Image file size in bytes
mode	Image color mode (ex: RGB, greyscale)
image_width	Image width in pixels
image_height	Image height in pixels
aspect_ratio	Ratio of image width to image height

Table I: Image metadata fields.

The dataset consists of two primary components: a collection of mineral images and a corresponding metadata structure. The image dataset contains JPEG photographs organized into two classes: silver-bearing minerals (positive class) and non-silver minerals (negative class), supporting a binary classification framework. Each image is paired with metadata that includes the mineral name, assigned class label, and extracted image characteristics such as dimensions, file size, and format. The metadata are stored in both CSV and JSON formats to ensure compatibility with Python-based machine learning workflows and to provide flexible access for analysis and validation. The silver image class comprises both native silver specimens and silver-bearing minerals. The non-silver class consisted of approximately 70% “hard negatives” and 30% “easy negatives”. Hard negatives are non-silver images that are visually similar to the silver class and difficult to distinguish, while easy negatives are clearly different and easy to classify. The silver class mineral names are silver, Argentum, Chlorargyrite, Proustite, Pyrargyrite, Stephanite, Polybasite, Pearceite, Miargyrite, and Freibergite. The “hard negative” minerals are Galena, Pyrite, Hematite, Graphite, Stibnite, Sphalerite, Chalcopyrite, and Molybdenite with “easy negatives” being all other minerals available from Mindat. This design choice was intended to improve model robustness by including visually similar mineral samples, thereby reducing both false positives (misclassifying non-silver as silver) and false negatives (misclassifying silver-bearing minerals as non-silver). All data and metadata were stored on Google Drive, and a link to that drive is located on the team’s Github README file.

Mindat does not support image collection through their API, so a website scraping technique was used to collect the images in the dataset. A large number of them were corrupted or blocked, so those were omitted from the set. Additionally, many of the images labeled as one of the types of silver consisted of more than just silver in the image (i.e. an image of a silver mine), so the collection was altered to only grab images with labels of three or fewer words. These filtering techniques resulted in a clean dataset of only minerals. Initial training tests used a 5-1 ratio of non-silver to silver images, but poor results resulted in a switch to a 1-1 ratio which was used for the remainder of project development and model testing.

II. Data Analysis (12%)

The two research questions explored in this project are as follows: (1) Can machine vision be used to accurately detect instances of silver? And (2) Does a Convolutional Neural Network (CNN) or a Vision Transformer (ViT) perform better on a relatively small image set, where a majority of the non-silver class consists of minerals similar in look to silver? The hypothesized answers to these research questions were (1) yes and (2) CNNs would prove more accurate results due to the smaller

dataset size. To find actual answers to these questions, five different models were trained on the same set of images and compared based on their accuracy and recall scores: three CNNs and two ViTs. The specific models used were ResNet-50 (RN50), DenseNet-121 (DN121), EfficientNet-B0 (EN-B0), Baseline Vision Transformer (ViT-B/16), and Data-efficient Image Transformer (DeiT-S). The RN50 is a deep CNN which uses residual connections to address the vanishing gradient problem, and is well-known for its strong performance and stability in image classification tasks. DN121 is a densely connected CNN, showing strong performance in tasks requiring fine-grained feature extraction. EN-B0, the final CNN compared, has shown to be effective for moderately sized datasets and resource-constrained environments by uniformly scaling network depth, width, and resolution. ViT-B/16, the basic Vision Transformer, widely demonstrates strong performance on large-scale datasets by dividing images into fixed-size patches and process them using self-attention mechanisms to capture global relationships. This model often requires substantial data and/or pretraining to reach optimal performance, so the DeiT-S (which incorporates knowledge distillation) was used to observe its performance on a smaller dataset.

Evaluation metrics were, as is standard, the F1 score, but also another metric: recall. Recall was chosen in addition to F1 score as an evaluation metric because, in this application, the priority is on accurately detecting all instances of silver, and false positives are not as much of a concern as missed instances of the target class. Various classification thresholds on the test set were compared to find the ideal threshold for accurate classification, with the understanding that lower thresholds suggest less confident models. The structure of the project was broken into five individual Jupyter notebook files created and run in Google Colab. Their descriptions are as follows:

a. `create_data.ipynb`

This notebook performs automated data acquisition to produce images for the silver and non-silver classes. It reads the Mindat image URL list (stored as both a CSV and a JSON), filters minerals into silver and non-silver categories, and programmatically downloads available JPEG images. JSON and CSV were chosen because they are widely supported and allow interoperability and ease of use. This script includes error handling, logging, and class-balancing logic, including hard and easy negatives, to construct the final dataset. The images are stored in a Google Drive which is linked to the Google Colab Jupyter notebook and downloaded from there.

b. `data_exploration.ipynb`

This notebook explores the silver and non-silver dataset by generating distribution plots that show class balance, mineral types, and image characteristics such as width, aspect ratio, brightness, and contrast. It then prepares the dataset for machine learning by loading and formatting the images in Keras for CNN training. Distribution plots for the silver and non-silver classes were generated, as

shown in Figures 1 and 2, respectively. Other visualizations include image width distribution (Figure 3), aspect ratio distribution (Figure 4), image brightness distribution (Figure 5), and image contrast distribution (Figure 6).

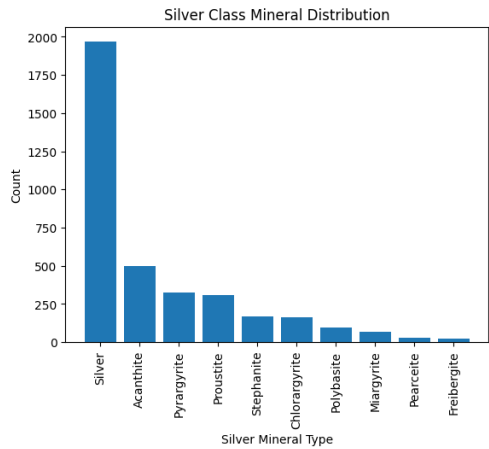


Figure 1: Silver class distribution

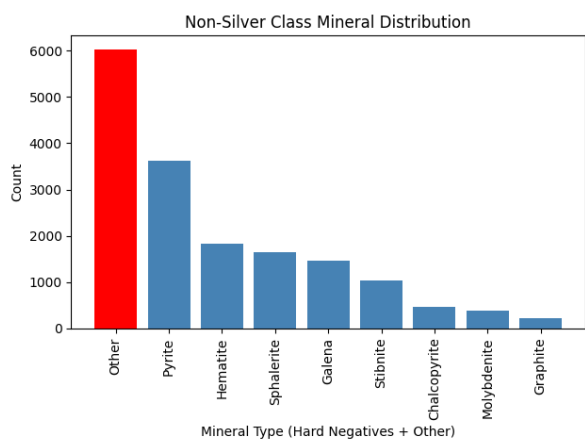


Figure 2: Non-silver class distribution

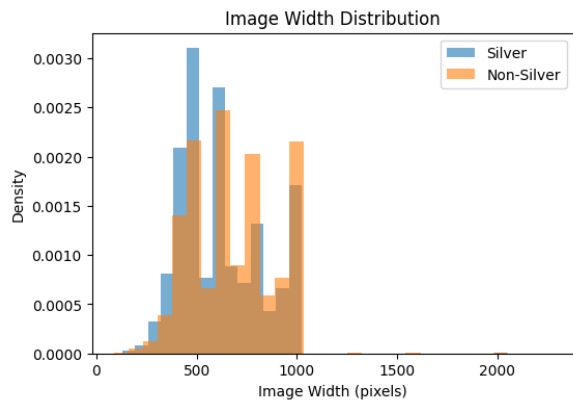


Figure 3: Class-wise image width distribution

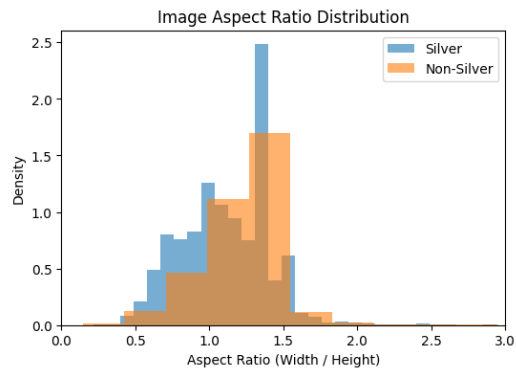


Figure 4: Class-wise aspect ratio distribution

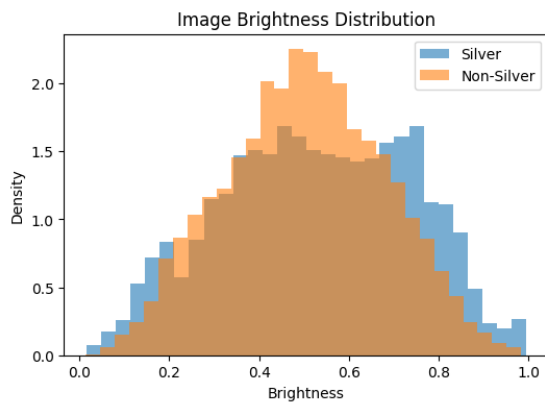


Figure 5: Class-wise image brightness distribution

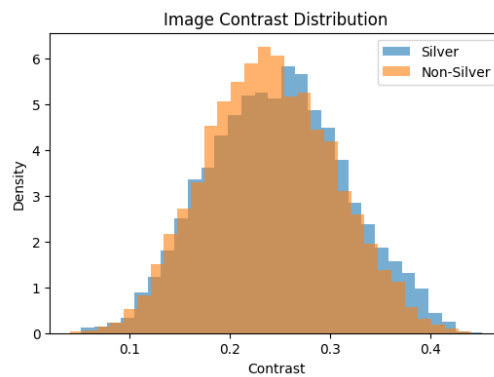


Figure 6: Class-wise image contrast distribution

The distribution graphs show a typical bell curve shape, suggesting potential higher model performance on images with consistent lighting but reliable training results on images on either extreme as well. These initial observations were encouraging, supporting the hypothesized answer to whether machine vision models are a viable tool in silver detection applications.

c. machine_learning_models.ipynb

This notebook trains the five machine learning models to classify mineral images as silver or non-silver using balanced datasets and structured preprocessing. The models include three CNN architectures, which are ResNet 50, DenseNet 121, and EfficientNet B0. They also include two transformer models, which are ViT Base Patch16 224 and DeiT Small Patch16 224. Each model is fine tuned and evaluated using standard metrics with threshold optimization to improve overall performance and maximize silver detection. The results are presented with graphs and performance metrics to support clear interpretation and comparison of model behavior. Table II shows results of all five compared models, including the best classification threshold and the recall and F1 scores at that threshold.

TABLE II: Model results comparison

Model Name \ Metric	Best Threshold	Recall	F1
RN50	0.35	0.8712	0.8503
DN121	0.50	0.8712	0.8657
EN-B0	0.22	0.9200	0.8600
ViT-B/16	0.37	0.8800	0.8900
DeiT-S	0.63	0.9068	0.8803

These results answered the initial research questions as (1) yes, and (2) the data-efficient Vision Transformer showed the best F1 score (Figure 7) while the Efficient Net CNN showed the best recall score (Figure 8) for optimized thresholds.

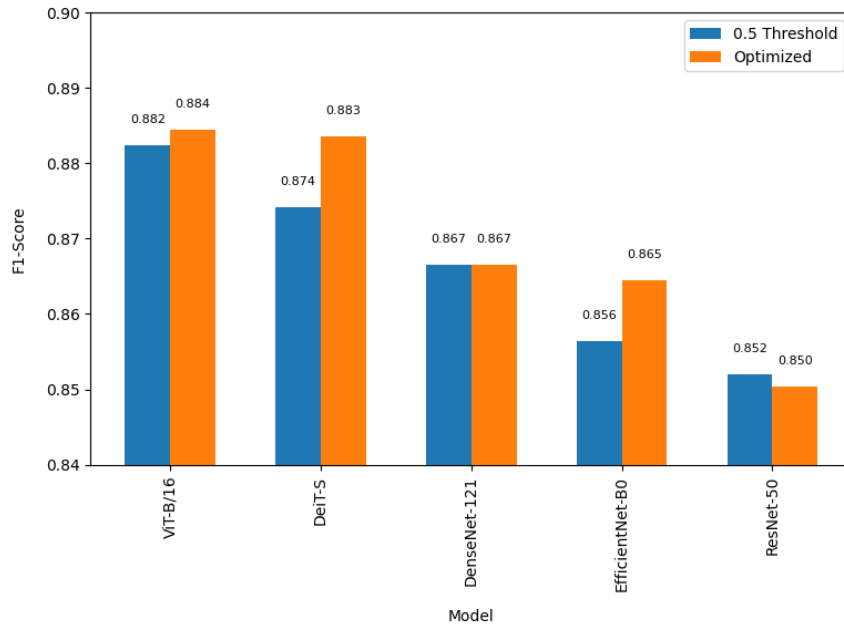


Figure 7: F1 score comparison for all models.

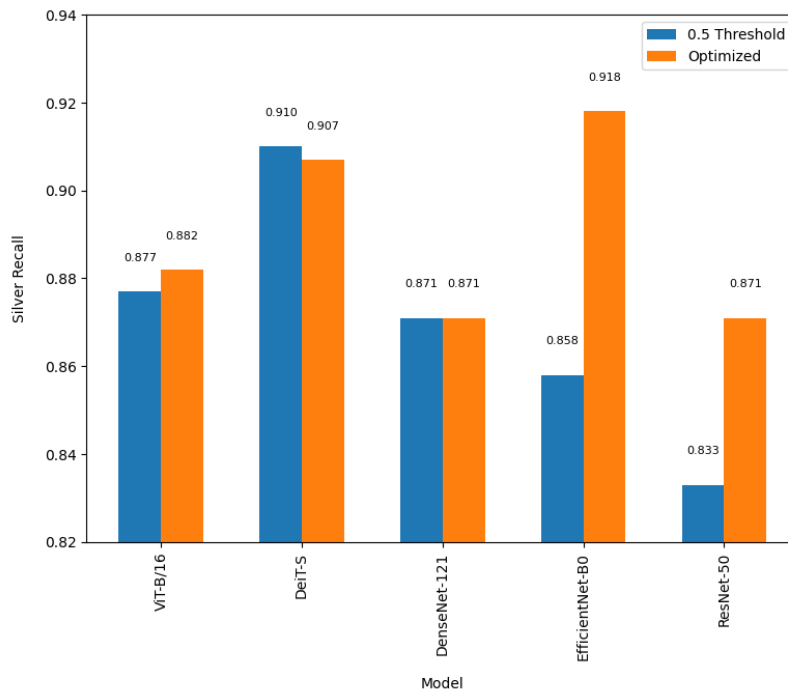


Figure 8: Recall score comparison for all models.

d. metadata.ipynb

This notebook builds and validates metadata for silver and non-silver image datasets used in training machine learning models. It verifies image files and extracts key metadata, including format, file size,

color mode, dimensions, and aspect ratio. All metadata is organized and exported to JSON and CSV formats to support dataset validation and downstream machine learning workflows.

e. `utility_JSON_conversion.ipynb`

This notebook provides a function to convert CSV files into JSON format. This notebook a conversion function use to generate a JSON file from CSV file.

III. Presentation/ Visualization (6%)

Figures 1 through 6 show the metadata and image properties of the images in the dataset. A sample image is shown in Figure 9. Section II c of this report contains a further in-depth analysis of model results, but the answers to the initial research questions can be summarized as (1) yes, and (2) the data-efficient Vision Transformer showed the best F1 score (Figure 7) while the Efficient Net CNN showed the best recall score (Figure 8) for optimized thresholds.



Figure 9: Sample images from the finalized, cleaned dataset.

Table I denotes all metadata collected for the silver image dataset. Both CSV and JSON formats store corresponding image data, where element 0 represents image_0 and element 1 represents image_1, continuing sequentially through the dataset. Metadata for silver images is stored in `silver_metadata.csv` and `silver_metadata.json`, while metadata for non-silver images is stored in `non_silver_metadata.csv` and `non_silver_metadata.json`. The `metadata/` folder in the Github repository contains structured information for all images in both the original and cleaned datasets.

The visualizations and Machine Learning allowed for the answering of the initial two research questions. Mindat made data access simple, even though we had to scrape the site itself for image data. Additionally, the access of these images allowed for the development of the experiments and tests which answered initial research questions. The second initial hypothesis was that CNNs would outperform ViTs, because typical ViTs require large amounts of data that we did not have. However, with the use of pretraining, the ViTs ended up outperforming the CNNs in both F1 and recall in the end, especially when comparing on the basis of non-optimized thresholds. An additional hypothesis leading into future work is that with no transfer learning, the CNNs would actually be the better choice for this use case and dataset size.

IV. Data Management Plan

The overall data management plan for this project for each of the 9 categories of data management are as follows:

- a. *Creation of logical collections:* The data was organized into logical collections based on mineral property (i.e. silver or non-silver) with additional grouping in the non-silver class by minerals deemed similar in look to silver minerals. The silver class mineral names are silver, Argentum, Chlorargyrite, Proustite, Pyrargyrite, Stephanite, Polybasite, Pearceite, Miargyrite, and Freibergite. The “hard negative” minerals are Galena, Pyrite, Hematite, Graphite, Stibnite, Sphalerite, Chalcopyrite, and Molybdenite with “easy negatives” being all other minerals available from Mindat.
- b. *Physical data handling:* All images come directly from the Mindat site. The URLs for accessing them have been scraped; they are downloaded as part of the project. The collection of URLs is stored inside of a CSV file. The collected data will be stored digitally as structured comma-separated values (CSV) files on a local computer and backed up using cloud storage (a GitHub repository). Each dataset version will be clearly labeled with dates and sources.
- c. *Interoperability support:* Metadata pertaining to images is stored in both a CSV and JSON format. These are both very widely supported and allow interoperability and ease of use.
- d. *Security support:* Since the data contains no personal or sensitive information, minimal security measures are required. The GitHub page is public, linked to at the bottom of this report.
- e. *Data ownership:* The original mineral image data is owned by Mindat and subject to its licensing terms. The compiled and curated dataset created for this assignment will be owned by the student, properly crediting Mindat as the original data source.
- f. *Metadata collection, management, and access:* Metadata was collected to document the source of the data, date of access, data preprocessing steps, and information pertinent to each image.

Table I documents these field names specifically. This metadata is stored in the metadata folder accompanying the dataset in the GitHub repository.

- g. *Persistence*: Metadata is hosted on GitHub. Fetched images are hosted on Google Drive, linked to within the repository. The process of fetching the images and the source code will be hosted on GitHub. Thus, an updated set of images can be fetched by future researchers or users at any time.
- h. *Discovery*: Data will be freely discoverable, public, and shared on GitHub. Clear file names, descriptive metadata, and keywords such as “silver images” and “image mineral dataset” used in the repository README files will support discovery. Documentation and a shareable GitHub repository will allow for easy understanding, sharing, and discovery.
- i. *Data dissemination and publication*: There will be no advertisement(s) nor guarantee of updates. However, the original Mindat dataset and front-end access point will be linked, and documentation will include appropriate citations and acknowledgments to Mindat.

V. Appendices

Appendix A: GitHub repository

<https://github.com/JMGillum/CS4621-Silver-Detection/tree/main>